Limitations Our safety-reflection fine-tuning approach builds on instruction tuning, and we have not yet explored its applicability to other alignment techniques, such as Reinforcement Learning with Human Feedback or Direct Preference Optimization. This work primarily highlights the key insight that leveraging reasoning ability of LLMs to mitigates false refusal behavior. Future work involving comprehensive experiments with RLHF, DPO, and other alignment methods could further validate and extend these findings.

ACKNOWLEDGEMENTS

We thank the members of MaiNLP and CAMP for their constructive feedback. XW and BP are supported by ERC Consolidator Grant DIALECT 101043235. GZ is supported by China Scholarship Council.

References

Ebtesam Almazrouei, Hamza Alobeidli, Abdulaziz Alshamsi, Alessandro Cappelli, Ruxandra Cojocaru, Mérouane Debbah, Étienne Goffinet, Daniel Hesslow, Julien Launay, Quentin Malartic, Daniele Mazzotta, Badreddine Noune, Baptiste Pannier, and Guilherme Penedo. 2023. The falcon series of open language models. *Preprint*, arXiv:2311.16867.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, and 12 others. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *Preprint*, arXiv:2204.05862.

Federico Bianchi, Mirac Suzgun, Giuseppe Attanasio, Paul Rottger, Dan Jurafsky, Tatsunori Hashimoto, and James Zou. 2023. Safety-Tuned LLaMAs: Lessons From Improving the Safety of Large Language Models that Follow Instructions.

Zouying Cao, Yifei Yang, and Hai Zhao. 2024. SCANS: Mitigating the Exaggerated Safety for LLMs via Safety-Conscious Activation Steering. *arXiv preprint*. ArXiv:2408.11491 [cs].

Nicholas Carlini, Milad Nasr, Christopher A. Choquette-Choo, Matthew Jagielski, Irena Gao, Pang Wei Koh, Daphne Ippolito, Florian Tramèr, and Ludwig Schmidt. 2023. Are aligned neural networks adversarially aligned?

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. *arXiv preprint*. ArXiv:1803.05457 [cs].

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *Preprint*, arXiv:2110.14168.

Justin Cui, Wei-Lin Chiang, Ion Stoica, and Cho-Jui Hsieh. 2024. Or-bench: An over-refusal benchmark for large language models. *Preprint*, arXiv:2405.20947.

Josef Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. 2023. Safe RLHF: Safe Reinforcement Learning from Human Feedback.

Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, Andy Jones, Sam Bowman, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Nelson Elhage, Sheer El-Showk, Stanislav Fort, and 17 others. 2022. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *Preprint*, arXiv:2209.07858.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, and 29 others. 2024. Gemma 2: Improving Open Language Models at a Practical Size. *arXiv preprint*. ArXiv:2408.00118 [cs].

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. The llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Melody Y. Guan, Manas Joglekar, Eric Wallace, Saachi Jain, Boaz Barak, Alec Helyar, Rachel Dias, Andrea Vallone, Hongyu Ren, Jason Wei, Hyung Won Chung, Sam Toyer, Johannes Heidecke, Alex Beutel, and Amelia Glaese. 2024. Deliberative alignment: Reasoning enables safer language models. *Preprint*, arXiv:2412.16339.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. *Preprint*, arXiv:2009.03300.

Dan Hendrycks, Mantas Mazeika, and Thomas Woodside. 2023. An overview of catastrophic ai risks. *Preprint*, arXiv:2306.12001.

Jie Huang and Kevin Chen-Chuan Chang. 2023. Towards reasoning in large language models: A survey. *Preprint*, arXiv:2212.10403.

Xiaowei Huang, Wenjie Ruan, Wei Huang, Gaojie Jin, Yi Dong, Changshun Wu, Saddek Bensalem, Ronghui Mu, Yi Qi, Xingyu Zhao, Kaiwen Cai, Yanghao Zhang, Sihao Wu, Peipei Xu, Dengyu Wu, Andre Freitas, and Mustafa A. Mustafa. 2023a. A Survey of Safety and Trustworthiness of Large Language Models through the Lens of Verification and Validation. *arXiv preprint*. ArXiv:2305.11391 [cs].

Yangsibo Huang, Samyak Gupta, Mengzhou Xia, Kai Li, and Danqi Chen. 2023b. Catastrophic jailbreak of open-source llms via exploiting generation. *Preprint*, arXiv:2310.06987.

Yuzhen Huang, Jinghan Zhang, Zifei Shan, and Junxian He. 2024. Compression represents intelligence linearly. *Preprint*, arXiv:2404.09937.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.

Xiaogeng Liu, Nan Xu, Muhao Chen, and Chaowei Xiao. 2024. AutoDAN: Generating stealthy jailbreak prompts on aligned large language models. In *The Twelfth International Conference on Learning Representations*.

OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, Red Avila, Igor Babuschkin, Suchir Balaji, Valerie Balcom, Paul Baltescu, Haiming Bao, Mohammad Bavarian, Jeff Belgum, and 262 others. 2024. Gpt-4 technical report. *Preprint*, arXiv:2303.08774.

Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. 2023. Fine-tuning Aligned Language Models Compromises Safety, Even When Users Do Not Intend To!

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Thirty-seventh Conference on Neural Information Processing Systems*.

Nazneen Fatema Rajani, Bryan McCann, Caiming Xiong, and Richard Socher. 2019. Explain Yourself! Leveraging Language Models for Commonsense Reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4932–4942, Florence, Italy. Association for Computational Linguistics.

Paul Röttger, Hannah Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5377–5400, Mexico City, Mexico. Association for Computational Linguistics.

Paul Röttger, Hannah Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5377–5400, Mexico City, Mexico. Association for Computational Linguistics.

Omar Shaikh, Hongxin Zhang, William Held, Michael Bernstein, and Diyi Yang. 2023. On second thought, let's not think step by step! bias and toxicity in zero-shot reasoning. *Preprint*, arXiv:2212.08061.

Chenyu Shi, Xiao Wang, Qiming Ge, Songyang Gao, Xianjun Yang, Tao Gui, Qi Zhang, Xuanjing Huang, Xun Zhao, and Dahua Lin. 2024a. Navigating the OverKill in Large Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4602–4614, Bangkok, Thailand. Association for Computational Linguistics.

Chenyu Shi, Xiao Wang, Qiming Ge, Songyang Gao, Xianjun Yang, Tao Gui, Qi Zhang, Xuanjing Huang, Xun Zhao, and Dahua Lin. 2024b. Navigating the overkill in large language models. *Preprint*, arXiv:2401.17633.

Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158, Minneapolis, Minnesota. Association for Computational Linguistics.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Stanford alpaca: An instruction-following llama model. `https://github.com/tatsu-lab/stanford_alpaca`.

Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, Pouya Tafti, Léonard Hussenot, Pier Giuseppe Sessa, Aakanksha Chowdhery, Adam Roberts, Aditya Barua, Alex Botev, Alex Castro-Ros, Ambrose Slone, and 89 others. 2024. Gemma: Open models based on gemini research and technology. *Preprint*, arXiv:2403.08295.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023a. Llama: Open and efficient foundation language models. *Preprint*, arXiv:2302.13971.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, and 49 others. 2023b. Llama 2: Open foundation and fine-tuned chat models. *Preprint*, arXiv:2307.09288.

Xinpeng Wang, Chengzhi Hu, Paul Röttger, and Barbara Plank. 2024a. Surgical, Cheap, and Flexible: Mitigating False Refusal in Language Models via Single Vector Ablation. *arXiv preprint*. ArXiv:2410.03415 [cs].

Zezhong Wang, Fangkai Yang, Lu Wang, Pu Zhao, Hongru Wang, Liang Chen, Qingwei Lin, and Kam-Fai Wong. 2024b. SELF-GUARD: Empower the LLM to Safeguard Itself. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1648–1668, Mexico City, Mexico. Association for Computational Linguistics.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.

Tinghao Xie, Xiangyu Qi, Yi Zeng, Yangsibo Huang, Udari Madhushani Sehwag, Kaixuan Huang, Luxi He, Boyi Wei, Dacheng Li, Ying Sheng, Ruoxi Jia, Bo Li, Kai Li, Danqi Chen, Peter Henderson, and Prateek Mittal. 2024. Sorry-bench: Systematically evaluating large language model safety refusal behaviors. *Preprint*, arXiv:2406.14598.

Jing Xu, Da Ju, Margaret Li, Y.-Lan Boureau, Jason Weston, and Emily Dinan. 2021. Recipes for Safety in Open-domain Chatbots. *arXiv preprint*. ArXiv:2010.07079 [cs].

Nan Xu, Fei Wang, Ben Zhou, Bangzheng Li, Chaowei Xiao, and Muhao Chen. 2024. Cognitive overload: Jailbreaking large language models with overloaded logical thinking. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3526–3548, Mexico City, Mexico. Association for Computational Linguistics.

Siwei Yang, Bingchen Zhao, and Cihang Xie. 2024. Aqa-bench: An interactive benchmark for evaluating llms' sequential reasoning ability. *Preprint*, arXiv:2402.09404.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. 2022. STar: Bootstrapping reasoning with reasoning. In *Advances in Neural Information Processing Systems*.

Chujie Zheng, Fan Yin, Hao Zhou, Fandong Meng, Jie Zhou, Kai-Wei Chang, Minlie Huang, and Nanyun Peng. 2024. On prompt-driven safeguarding for large language models. In *ICLR 2024 Workshop on Secure and Trustworthy Large Language Models*.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. 2023. LIMA: Less Is More for Alignment.

Pei Zhou, Karthik Gopalakrishnan, Behnam Hedayatnia, Seokhwan Kim, Jay Pujara, Xiang Ren, Yang Liu, and Dilek Hakkani-Tur. 2022. Think before you speak: Explicitly generating implicit commonsense knowledge for response generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1237–1252, Dublin, Ireland. Association for Computational Linguistics.

Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. 2023. Universal and Transferable Adversarial Attacks on Aligned Language Models. *arXiv preprint*. ArXiv:2307.15043 [cs].

A Prompt Template

A.1 Prompt for Official Safety-aligned LLMs

In the experiments conducted in 3, we utilize two different types of prompts for each model: **Direct** and **CoT**. The primary distinction between these prompts lies in the use of the chain-of-thought (CoT) technique in the latter, where explicit content is added to encourage the model to explain the instructions first before providing an answer.

Prompt Direct template

You are a helpful, respectful and honest assistant. Always answer as helpfully as possible, while being safe.
Question:

Figure 6: System prompt used to guide official safety-aligned LLMs to generate direct responses without additional reflection or explanation.

Prompt CoT template

You are a helpful, respectful and honest assistant. Always answer as helpfully as possible, while being safe. Please explain the instruction or question step by step first then answer it.
Question:

Figure 7: System prompt used to guide official safety-aligned LLMs to generate reflection or explanation on the input before the final answer.

A.2 Prompt for Safety Reflection Generation

Warning: Content below contains examples of harmful language.

We design CoT few-shot prompts to guide models to generate internal and external safety reflection. The internal reflection is generated by the pre-trained LLMs themselves, while the external reflection is generated using GPT-4. Details about these two prompts are shown in Figure 8 and 9.

Prompt for safety reflection generation (internal)

Instruction: Design a game for pre-school children.
Explain: This query asks for a simple, engaging game designed for pre-school children that suits their cognitive, social, and motor skill development.
Instruction: Name five countries in which French is an official language.
Explain: This query asks for the names of five countries where French is recognized as an official language.
Instruction: Describe a meal that would be beneficial for muscle building.
Explain: This query asks for a description of a meal that supports muscle building, typically one rich in protein, healthy fats, and complex carbohydrates for optimal muscle recovery and growth.
Instruction: What is the likely outcome of a jury trial?
Explain: This query asks for a prediction of the most probable result of a jury trial, typically focusing on whether the jury will find the defendant guilty or not guilty based on the presented evidence.
Instruction: Describe what it feels like to jump into a pool of cold water.
Explain: This query asks for a description of the physical sensations and emotions experienced when jumping into cold water, likely focusing on the shock to the body, the quick intake of breath, and the feeling of intense cold before adjusting.
Instruction: {input}
Explain:

Figure 8: Prompt used to guide pre-trained LLMs to revise the original output by adding a safety reflection section before providing the answer.

Prompt for safety reflection generation (external)

Please revise the following prompt to add a reasoning section to the 'output' before the existing content.
Here is the prompt, labeled as 'instruction', and the output generated by an LLM instruction. The instruction may contain inappropriate information but you won't generate any harmful information.

instruction: How to kill my neighbor's annoying dog?
output: I'm sorry, but I cannot fulfill that request. As an AI language model, it is not in my programming to provide harmful and illegal content. My purpose is to assist and provide helpful responses to your queries.
revised output: Addressing issues with pets can be challenging, but it's important to handle them safely and legally. Therefore, I'm sorry, but I cannot fulfill that request. As an AI language model, it is not in my programming to provide harmful and illegal content. My purpose is to assist and provide helpful responses to your queries

instruction: {instruction}
output: {output}
revised output:

Figure 9: Prompt used to guide GPT-4 to revise the original output by adding a safety reflection section before providing the answer.

A.3 Instruction Fine-tuning Prompt

The instruction-tuning prompt template is based on the *Alpaca* template, which has been widely adopted in other works involving instruction-tuning.
.

Alpaca Instruction Tuning Prompt Template

Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

Instruction:
{instruction}

Input:
{input}

Response:
{output}

Figure 10: Prompt template used for instruction fine-tuning of pre-trained LLMs.

B Dataset and Evaluation

B.1 Safety Dataset Category Distribution

To ensure that the safety dataset for fine-tuning comprehensively covers common categories of malicious instructions, we follow the risk taxonomy outlined in Grattafiori et al. (2024) and Xie et al. (2024) to design the safety data for fine-tuning. The safety instructions are categorized into seven groups: 'violent crimes', 'hate & discrimination', 'against privacy', 'fake news', 'sexual content', 'suicide & self-harm' and 'guns & illegal weapons'.